

OAC RICO FIELD BEAN

OAC Rico, a backcross-derived anthracnose-resistant cultivar of field bean (*Phaseolus vulgaris* L.) similar in agronomic performance to Ex Rico 23, was developed by the Crop Science Department, University of Guelph. OAC Rico carries the *Are* gene that confers resistance to alpha, beta, delta, epsilon, gamma, and lambda races of anthracnose (*Colletotrichum lindemuthianum* (Sacc. and Magn.) Briosi and Cav.).

Key words: Cultivar description, anthracnose resistant, bean (white)

[Le haricot blanc OAC Rico.]

Titre abrégé: Haricot blanc OAC Rico.

OAC Rico est un cultivar de haricot blanc (*Phaseolus vulgaris* L.) résistant à l'anthracnose obtenu par rétrocroisement. Mis au point par le Crop Science Department de l'université de Guelph, ce cultivar présente des qualités agronomiques semblables à celles de Ex Rico 23. Il contient le gène *Are* qui confère la résistance aux races alpha, beta, delta, epsilon, gamma et lambda de l'anthracnose (*Colletotrichum lindemuthianum* (Sacc. et Magn.) Briosi et Cav.).

Mots clés: Description de cultivar, résistance à l'anthracnose, haricot blanc

Bean anthracnose, caused by *Colletotrichum lindemuthianum* (Sacc. and Magn.) Briosi and Cav., has not caused losses to white bean producers in Canada and the U.S.A. for the past two decades because commonly grown cultivars were resistant to the previously prevalent races (alpha, beta and gamma). An outbreak of anthracnose in southern Ontario during 1976 was caused largely by the delta race and to a lesser extent by the lambda race, both new to Canada. A backcross breeding program was initiated in 1977 to improve the anthracnose resistance of Ex Rico 23.

License no. 2326 was issued for cv. OAC Rico on 2 May 1983 by the Plant Health and Plant Products Directorate, Agriculture Canada. This paper describes the origin, characteristics and performance of this cultivar.

Pedigree and Breeding

OAC Rico originated from a cross and subsequent backcrosses of Ex Rico 23 (recurrent parent) × Narda (donor parent used as
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the source of *Are* gene for resistance to several races of anthracnose). OAC Rico represents homozygous (*Are/Are*) progeny of the sixth backcross (BC₆). Individual backcross families were purified and bulked as OAC 81-4 (also designated as Ex Rico-ARE in some cases).

Agronomic Performance

OAC Rico has been tested in Ontario Field Bean Variety Trials since 1981. Cooperative trials were conducted by the Colleges of Agricultural Technology at Ridgetown and Centralia; Crop Science Dep., University of Guelph and Agriculture Canada Research Station, Harrow. Over the 2-yr period 1981–1982, OAC Rico averaged 1 day later in maturity than Ex Rico 23, and yielded about 1% less than Ex Rico 23 (Table 1). It is adapted to the warmer part of the main field bean growing area in southern Ontario with at least 2800 corn heat units.

Description

OAC Rico is a viney-bush (determinate)

plant type with characteristics very similar to Ex Rico 23. OAC Rico has short runners (guides) in some environments and has white flowers with a large portion of the flowers placed within the leaf canopy. OAC Rico has creamy buff pods, some of which develop faint purple stripes in certain environments. It has white seeds similar in size and shape to Ex Rico 23 and typical of white beans, navy beans or Michigan pea beans. OAC Rico also has a tap root similar to Ex Rico 23.

Quality

Taste panel evaluations conducted by the Agriculture Canada Research Station, Harrow, Ontario (Table 1) and by Libby, McNeill, and Libby Ltd., Chatham, Ontario, rated OAC Rico as acceptable and comparable with the check cultivar Ex Rico 23 in processed (canned) products. Protein

content was also similar to the check cultivars. Seed size and shape are acceptable to the industry.

Disease Reaction

OAC Rico is resistant to races 1 and 15 of bean common mosaic virus, and resistant to alpha, beta, delta, epsilon, gamma, and lambda races of anthracnose (Table 2). It is distinctly more field-tolerant to white mold (*Sclerotinia sclerotiorum* (Lib.) DeBary) than other cultivars grown in Ontario except for Ex Rico 23 (Table 1).

Pedigreed Seed Stocks

Breeder seed was released through the Ontario Seed Stock Distribution Committee of the Canadian Seed Growers' Association. The Crop Science Department, University of Guelph will maintain breeder seed.

Table 1. Average performance of OAC Rico and check cultivars in 12 Ontario Field Bean Variety Trials grown at Brussels, Centralia, Dresden, Elora, Kenwood, Kippen and Ridgeway, 1981–1982

Cultivar	Yield (tonnes/ha)	Maturity (days)	Seed weight (g/100 seeds)	Cooking quality score†	White mold (% of plants infected)‡
OAC Rico	2.78	100	20.2	2.5	24
Ex Rico 23	2.82	99	20.5	2.7	20
Fleetwood	2.84	102	19.5	2.4	48
Kentwood	2.61	97	23.3	2.2	38
Seafarer	2.38	88	19.3	2.2	35

†Values shown are combined quality indices including factors for appearance, flavor, and texture of canned beans, based on samples from six trials (1981–1982) and rated from (1) poor to (5) excellent. Tests were conducted at the Agriculture Canada Research Station, Harrow, Ontario.

‡Mean of four trials infected with white mold: Dresden, Kerwood, and Ridgeway in 1981; Mitchell in 1982.

Table 2. Disease ratings of OAC Rico and Ex Rico 23 for races of anthracnose and Bean Common Mosaic Virus (BCMV)†

Cultivar	Anthracnose						BCMV	
	Alpha	Beta	Epsilon	Gamma	Lambda	Delta	1	15
	Disease severity‡							
OAC Rico	0	0	0	0	0	0	R§	R
Ex Rico 23	5	0	8	5	8	6	R	R

†Based on controlled environment test conducted at the Agriculture Canada Research Station, Harrow, Ontario in 1981–1982.

‡0 = resistant, 9 = highly susceptible.

§R = resistant, S = susceptible.

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